TRAIN MANAGEMENT SYSTEM

DBMS Case Study / Mini Project

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1. Introduction

The Train Management System is a database project built to help run an Indian railway network. It takes care of everything needed to keep trains running smoothly , from setting schedules and managing routes to customer be able to book tickets, assigning staff, and handling payments.

This case study shows how we planned the database step-by-step and built it using MySQL. It highlights important database skills, like organizing information cleanly and using built-in tools to automate tasks and keep the data secure.

Scope of the System

The system manages the following areas:

- **Trains:** Basic details, train types, and seating capacity.
- **Stations:** Information about cities, stations, and specific platforms.
- **Routes & Schedules:** Train timings and the stops they make along the way.
- **Staff:** Organizing employees by their specific jobs (like Drivers, Ticket Collectors, and Maintenance Staff).
- **Passengers & Bookings:** Creating user accounts and booking tickets.
- **Seat Reservations:** Picking specific coaches and seats for passengers.
- **Payments:** Processing and tracking ticket purchases.
- **Timetables:** A central hub to easily view all train arrivals and departures.

2. Requirement Identification

Train & Station Management

- The system shall store train details including Train Number (unique), Name, Type, and Capacity.
- Each train can have multiple schedules and follow a route with multiple station stops.

- The system shall maintain station information including Station ID, Name, City, and Platform Number.
- A station can be a stop for multiple trains (Many-to-Many relationship).

Schedule Management

- The system shall record schedules with Arrival Time, Departure Time, and Running Date.
- Each schedule is a weak entity that depends on a Train (1:N relationship).

Timetable Event (Aggregation)

- The system shall provide a Timetable Event entity that aggregates Train, Station, and Schedule.
- This aggregation entity serves as a unified reference point for Booking and Employee management.

Employee Management

- The system shall store Employee details: ID, Name, Salary, Aadhar Number, and Phone Number.
- Employees are classified via ISA hierarchy into: Driver (License No), Ticket Collector (Badge No, Assigned Zone), and Maintenance Staff (Shift Type, Trade Skill).
- Each employee is assigned to manage a Timetable Event (M:1 relationship).

Passenger Management

- The system shall store Passenger details: ID, Name, Age, Gender, Phone Number, and Aadhar Number.
- A passenger can make multiple bookings (1:N relationship).

Booking Management

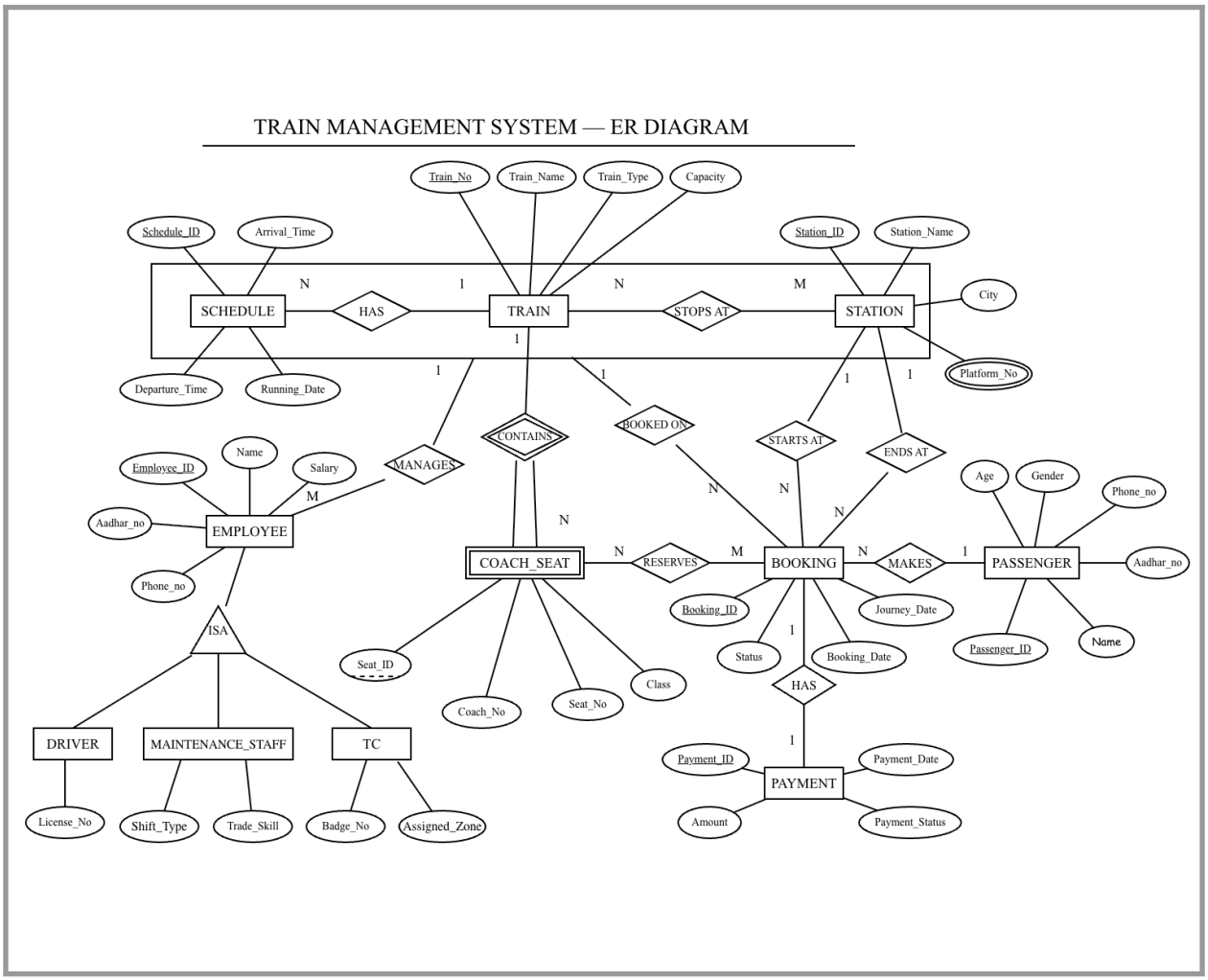
- The system shall create bookings with a unique Booking ID, linked to a Passenger, a Timetable Event, Start Station, End Station, Journey Date, Class, and Status.
- Booking status can be: Confirmed, Waiting, or Cancelled.

- The system shall manage Coach Seats (Coach No, Seat No, Class) for each train.
- Seats can be reserved against bookings (N:M relationship via BOOKING_SEAT).

- The system shall record payments with Payment ID, Amount, Payment Date, and Payment Status.
- Each booking has exactly one payment (1:1 relationship).

3. ER Diagram

The Entity-Relationship (ER) Diagram below represents the complete conceptual design of the Train Management System.

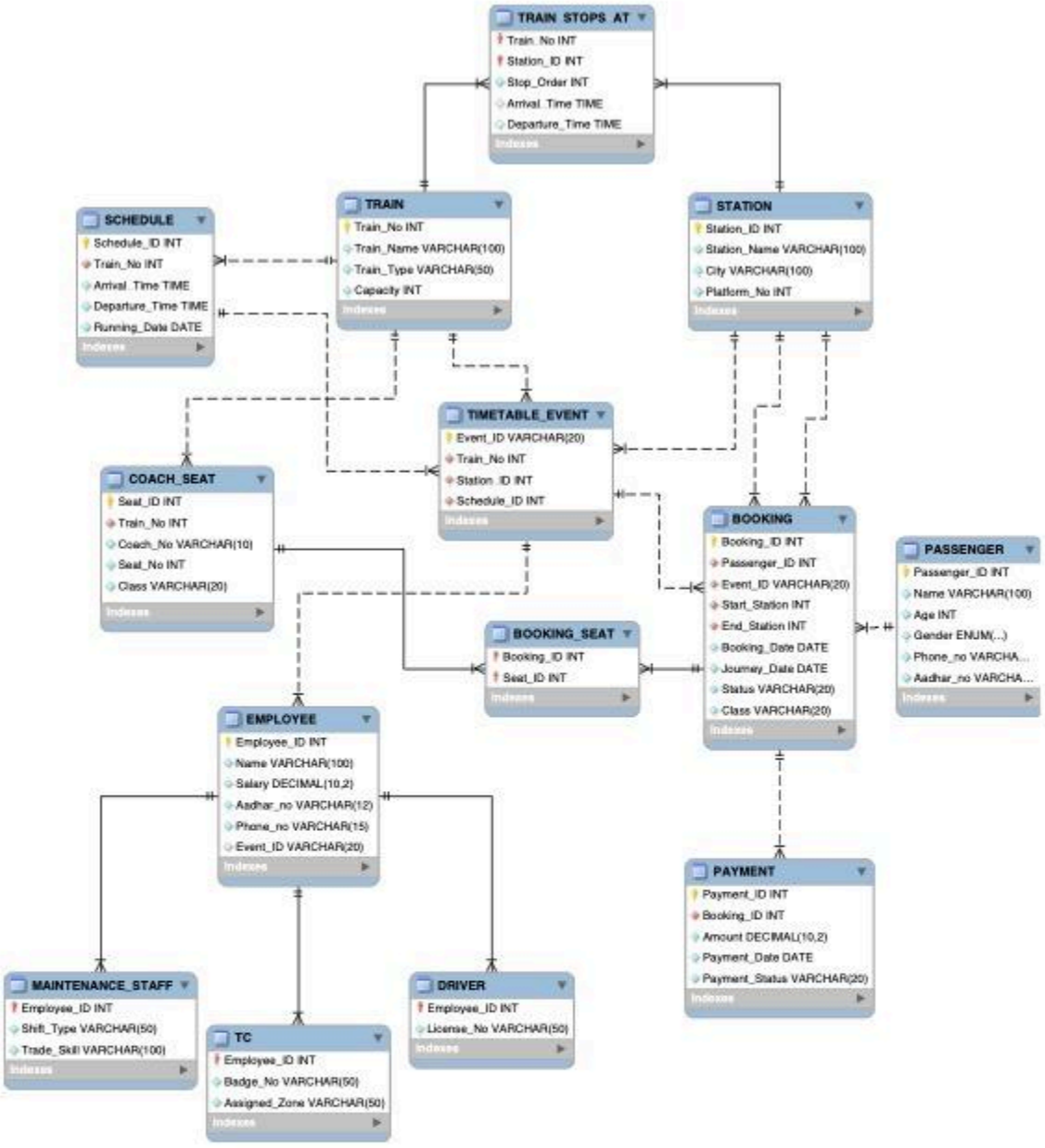


Key elements in the ER Diagram:

- **Strong Entities:** TRAIN, STATION, EMPLOYEE, PASSENGER, BOOKING, PAYMENT (single-bordered rectangles)
- **Weak Entities:** SCHEDULE, COACH_SEAT (double-bordered rectangles)
- **ISA Hierarchy:** EMPLOYEE → DRIVER, TC, MAINTENANCE_STAFF (triangle with "ISA" label)
- Aggregation: TIMETABLE_EVENT aggregates TRAIN, STATION, SCHEDULE (dashed rectangle)
- **Relationships:** HAS (1:N), STOPS AT (N:M), CONTAINS (1:N), MANAGES (M:1), MAKES (1:N), RESERVES (N:M), BOOKED ON (N:1), HAS (1:1 for Payment)

4. Relational Schema

The ER Diagram is mapped to the following relational schema. Each entity is converted to a table, M:N relationships become junction tables, weak entities include the owner's primary key as a foreign key, and the ISA hierarchy is implemented using separate tables with shared primary keys.



Schema Notation

Primary Keys are shown in bold and underlined. Foreign Keys are marked with (FK).

TRAIN (Train_No (PK), Train_Name, Train_Type, Capacity)

STATION (Station_ID (PK), Station_Name, City, Platform_No)

SCHEDULE (Schedule_ID (PK), Train_No (FK), Arrival_Time, Departure_Time, Running_Date)

TRAIN_STOPS_AT (Train_No (PK, FK), Station_ID (PK, FK), Stop_Order, Arrival_Time, Departure_Time)

TIMETABLE_EVENT (Event_ID (PK), Train_No (FK), Station_ID (FK), Schedule_ID (FK))

EMPLOYEE (Employee_ID (PK), Name, Salary, Aadhar_no, Phone_no, Event_ID (FK))

DRIVER (Employee_ID (PK, FK), License_No)

TC (Employee_ID (PK, FK), Badge_No, Assigned_Zone)

MAINTENANCE_STAFF (Employee_ID (PK, FK), Shift_Type, Trade_Skill)

COACH_SEAT (Seat_ID (PK), Train_No (FK), Coach_No, Seat_No, Class)

PASSENGER (Passenger_ID (PK), Name, Age, Gender, Phone_no, Aadhar_no)

BOOKING (Booking_ID (PK), Passenger_ID (FK), Event_ID (FK), Start_Station (FK), End_Station (FK), Booking_Date, Journey_Date, Status, Class)

BOOKING_SEAT (Booking_ID (PK, FK), Seat_ID (PK, FK))

PAYMENT (Payment_ID (PK), Booking_ID (FK, UNIQUE), Amount, Payment_Date, Payment_Status)

5. Identification of Primary Keys and Foreign Keys

Primary Keys

Primary keys uniquely identify each record in a table. The following primary keys are defined:

Table	Primary Key	Type	Auto-Increment
TRAIN	Train_No	INT	No (user-defined)
STATION	Station_ID	INT	Yes
SCHEDULE	Schedule_ID	INT	Yes
TRAIN_STOPS_AT	(Train_No, Station_ID)	Composite	No
TIMETABLE_EVENT	Event_ID	VARCHAR(20)	No (user-defined)
EMPLOYEE	Employee_ID	INT	Yes
DRIVER	Employee_ID	INT	No (inherits from EMPLOYEE)
TC	Employee_ID	INT	No (inherits from EMPLOYEE)
MAINTENANCE_STAFF	Employee_ID	INT	No (inherits from EMPLOYEE)
COACH_SEAT	Seat_ID	INT	Yes
PASSENGER	Passenger_ID	INT	Yes
BOOKING	Booking_ID	INT	Yes
BOOKING_SEAT	(Booking_ID, Seat_ID)	Composite	No
PAYMENT	Payment_ID	INT	Yes

Foreign Keys

Foreign keys enforce referential integrity between related tables:

Table	Foreign Key Column	References	ON DELETE	ON UPDATE
SCHEDULE	Train_No	TRAIN(Train_No)	CASCADE	CASCADE
TRAIN_STOPS_AT	Train_No	TRAIN(Train_No)	CASCADE	CASCADE
TRAIN_STOPS_AT	Station_ID	STATION(Station_ID)	CASCADE	CASCADE
TIMETABLE_EVENT	Train_No	TRAIN(Train_No)	CASCADE	CASCADE
TIMETABLE_EVENT	Station_ID	STATION(Station_ID)	CASCADE	CASCADE
TIMETABLE_EVENT	Schedule_ID	SCHEDULE(Schedule_ID)	CASCADE	CASCADE
EMPLOYEE	Event_ID	TIMETABLE_EVENT(Event_ID)	SET NULL	CASCADE
DRIVER	Employee_ID	EMPLOYEE(Employee_ID)	CASCADE	CASCADE
TC	Employee_ID	EMPLOYEE(Employee_ID)	CASCADE	CASCADE
MAINTENANCE_STAFF	Employee_ID	EMPLOYEE(Employee_ID)	CASCADE	CASCADE
COACH_SEAT	Train_No	TRAIN(Train_No)	CASCADE	CASCADE
BOOKING	Passenger_ID	PASSENGER(Passenger_ID)	CASCADE	CASCADE

BOOKING	Event_ID	TIMETABLE_EVENT(Event_ID)	CASCADE	CASCADE
BOOKING	Start_Station	STATION(Station_ID)	CASCADE	CASCADE
BOOKING	End_Station	STATION(Station_ID)	CASCADE	CASCADE
BOOKING_SEAT	Booking_ID	BOOKING(Booking_ID)	CASCADE	CASCADE
BOOKING_SEAT	Seat_ID	COACH_SEAT(Seat_ID)	CASCADE	CASCADE
PAYMENT	Booking_ID	BOOKING(Booking_ID)	CASCADE	CASCADE

6. Appropriate Constraints

The database enforces data integrity through multiple types of constraints:

NOT NULL Constraints

Critical columns are marked NOT NULL to prevent missing data:

- TRAIN: Train_Name, Train_Type, Capacity
- STATION: Station_Name, City, Platform_No
- EMPLOYEE: Name, Salary, Aadhar_no, Phone_no
- PASSENGER: Name, Age, Gender, Phone_no, Aadhar_no
- BOOKING: Passenger_ID, Event_ID, Start_Station, End_Station, Booking_Date, Journey_Date, Class
- PAYMENT: Booking_ID, Amount, Payment_Date

UNIQUE Constraints

Table	Column(s)	Purpose
EMPLOYEE	Aadhar_no	Each employee has a unique Aadhar number
DRIVER	License_No	Each driver has a unique driving license
TC	Badge_No	Each ticket collector has a unique badge
PASSENGER	Aadhar_no	Each passenger has a unique Aadhar number
COACH_SEAT	(Train_No, Coach_No, Seat_No)	No duplicate seat on same train/coach
PAYMENT	Booking_ID	Enforces 1:1 relationship with BOOKING

CHECK Constraints

Table	Column	Constraint	Purpose
TRAIN	Capacity	Capacity > 0	Train must have positive capacity
STATION	Platform_No	Platform_No > 0	Valid platform number
EMPLOYEE	Salary	Salary > 0	Salary must be positive
PASSENGER	Age	0 < Age < 150	Valid human age range
PAYMENT	Amount	Amount > 0	Payment must be positive

DEFAULT Constraints

- BOOKING.Status defaults to 'Confirmed'
- PAYMENT.Payment_Status defaults to 'Pending'

ENUM Constraint

- PASSENGER.Gender is restricted to ENUM('Male', 'Female', 'Other')

Referential Integrity (CASCADE Rules)

All foreign keys use ON DELETE CASCADE and ON UPDATE CASCADE, except EMPLOYEE.Event_ID which uses ON DELETE SET NULL (so employees are not deleted when a timetable event is removed, they are simply unassigned).

7. Database Creation using MySQL

The database is created using the following MySQL commands:

GitHub link:

```sql
CREATE DATABASE TrainManagementSystem;
USE TrainManagementSystem;

-- 1. TRAIN Table
CREATE TABLE TRAIN (
    Train_No      INT          PRIMARY KEY,
    Train_Name    VARCHAR(100) NOT NULL,
    Train_Type    VARCHAR(50)  NOT NULL,
    Capacity      INT          NOT NULL CHECK (Capacity > 0)
);


-- 2. STATION Table
CREATE TABLE STATION (
    Station_ID    INT          PRIMARY KEY AUTO_INCREMENT,
    Station_Name  VARCHAR(100) NOT NULL,
    City          VARCHAR(100) NOT NULL,
    Platform_No   INT          NOT NULL CHECK (Platform_No > 0)
);


-- 3. SCHEDULE – Weak Entity (depends on TRAIN via HAS)
CREATE TABLE SCHEDULE (
    Schedule_ID    INT          PRIMARY KEY AUTO_INCREMENT,
    Train_No       INT          NOT NULL,
    Arrival_Time   TIME         NOT NULL,
    Departure_Time TIME         NOT NULL,
    Running_Date   DATE         NOT NULL,
    FOREIGN KEY (Train_No) REFERENCES TRAIN(Train_No)
```

```
30          ON DELETE CASCADE ON UPDATE CASCADE
31  );
32
33  -- 4. TRAIN_STOPS_AT — Junction table for N:M relationship
34  CREATE TABLE TRAIN_STOPS_AT (
35      Train_No      INT NOT NULL,
36      Station_ID    INT NOT NULL,
37      Stop_Order    INT NOT NULL,                    -- sequence of stop
38      Arrival_Time  TIME,
39      Departure_Time TIME,
40      PRIMARY KEY (Train_No, Station_ID),
41      FOREIGN KEY (Train_No)   REFERENCES TRAIN(Train_No)
42          ON DELETE CASCADE ON UPDATE CASCADE,
43      FOREIGN KEY (Station_ID) REFERENCES STATION(Station_ID)
44          ON DELETE CASCADE ON UPDATE CASCADE
45  );
46
47  -- 5. TIMETABLE_EVENT — Aggregation of TRAIN, STATION, SCHEDULE
48  CREATE TABLE TIMETABLE_EVENT (
49      Event_ID      VARCHAR(20)  PRIMARY KEY,
50      Train_No      INT          NOT NULL,
51      Station_ID    INT          NOT NULL,
52      Schedule_ID   INT          NOT NULL,
53      FOREIGN KEY (Train_No)    REFERENCES TRAIN(Train_No)
54          ON DELETE CASCADE ON UPDATE CASCADE,
55      FOREIGN KEY (Station_ID)  REFERENCES STATION(Station_ID)
56          ON DELETE CASCADE ON UPDATE CASCADE,
57      FOREIGN KEY (Schedule_ID) REFERENCES SCHEDULE(Schedule_ID)
58          ON DELETE CASCADE ON UPDATE CASCADE
```

```sql
);

-- 6. EMPLOYEE Table
CREATE TABLE EMPLOYEE (
    Employee_ID   INT            PRIMARY KEY AUTO_INCREMENT,
    Name          VARCHAR(100)   NOT NULL,
    Salary        DECIMAL(10,2)  NOT NULL CHECK (Salary > 0),
    Aadhar_no     VARCHAR(12)    NOT NULL UNIQUE,
    Phone_no      VARCHAR(15)    NOT NULL,
    Event_ID      VARCHAR(20),                      -- MANAGES aggregation (M:1)
    FOREIGN KEY (Event_ID) REFERENCES TIMETABLE_EVENT(Event_ID)
        ON DELETE SET NULL ON UPDATE CASCADE
);

-- 6. DRIVER – ISA subtype of EMPLOYEE
CREATE TABLE DRIVER (
    Employee_ID   INT          PRIMARY KEY,
    License_No    VARCHAR(50)  NOT NULL UNIQUE,
    FOREIGN KEY (Employee_ID) REFERENCES EMPLOYEE(Employee_ID)
        ON DELETE CASCADE ON UPDATE CASCADE
);

-- 7. TC (Ticket Collector) – ISA subtype of EMPLOYEE
CREATE TABLE TC (
    Employee_ID    INT          PRIMARY KEY,
    Badge_No       VARCHAR(50)  NOT NULL UNIQUE,
    Assigned_Zone  VARCHAR(50)  NOT NULL,
    FOREIGN KEY (Employee_ID) REFERENCES EMPLOYEE(Employee_ID)
        ON DELETE CASCADE ON UPDATE CASCADE
);

-- 8. MAINTENANCE_STAFF – ISA subtype of EMPLOYEE
CREATE TABLE MAINTENANCE_STAFF (
    Employee_ID    INT          PRIMARY KEY,
    Shift_Type     VARCHAR(50)  NOT NULL,          -- Day, Night, Rotational
    Trade_Skill    VARCHAR(100) NOT NULL,          -- Electrical, Mechanical, etc.
```

```sql
    FOREIGN KEY (Employee_ID) REFERENCES EMPLOYEE(Employee_ID)
        ON DELETE CASCADE ON UPDATE CASCADE
);

-- 9. COACH_SEAT – Weak Entity (depends on TRAIN via CONTAINS)
--    Seat_ID is derived from Coach_No (shown as dashed oval)
CREATE TABLE COACH_SEAT (
    Seat_ID       INT          PRIMARY KEY AUTO_INCREMENT,
    Train_No      INT          NOT NULL,
    Coach_No      VARCHAR(10)  NOT NULL,
    Seat_No       INT          NOT NULL,
    Class         VARCHAR(20)  NOT NULL,
    UNIQUE (Train_No, Coach_No, Seat_No),
    FOREIGN KEY (Train_No) REFERENCES TRAIN(Train_No)
        ON DELETE CASCADE ON UPDATE CASCADE
);

-- 10. PASSENGER Table
CREATE TABLE PASSENGER (
    Passenger_ID  INT          PRIMARY KEY AUTO_INCREMENT,
    Name          VARCHAR(100) NOT NULL,
    Age           INT          NOT NULL CHECK (Age > 0 AND Age < 150),
    Gender        ENUM('Male', 'Female', 'Other') NOT NULL,
    Phone_no      VARCHAR(15)  NOT NULL,
    Aadhar_no     VARCHAR(12)  NOT NULL UNIQUE
);


-- 11. BOOKING – Strong Entity
--     Relationships:
--       PASSENGER 1 ——— MAKES ——— N BOOKING
--       BOOKING   N ——— STARTS AT ——— 1 STATION
--       BOOKING   N ——— ENDS AT   ——— 1 STATION
--       BOOKING   N ——— BOOKED ON ——— 1 TIMETABLE_EVENT
CREATE TABLE BOOKING (
    Booking_ID     INT          PRIMARY KEY AUTO_INCREMENT,
```

```sql
    Booking_ID     INT          PRIMARY KEY AUTO_INCREMENT,
    Passenger_ID   INT          NOT NULL,
    Event_ID       VARCHAR(20)  NOT NULL,          -- BOOKED ON aggregation
    Start_Station  INT          NOT NULL,          -- STARTS AT
    End_Station    INT          NOT NULL,          -- ENDS AT
    Booking_Date   DATE         NOT NULL,
    Journey_Date   DATE         NOT NULL,
    Status         VARCHAR(20)  NOT NULL DEFAULT 'Confirmed',  -- Confirmed, Waiting, Cancelled
    Class          VARCHAR(20)  NOT NULL,
    FOREIGN KEY (Passenger_ID)  REFERENCES PASSENGER(Passenger_ID)
        ON DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN KEY (Event_ID)      REFERENCES TIMETABLE_EVENT(Event_ID)
        ON DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN KEY (Start_Station) REFERENCES STATION(Station_ID)
        ON DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN KEY (End_Station)   REFERENCES STATION(Station_ID)
        ON DELETE CASCADE ON UPDATE CASCADE
);

-- 12. BOOKING_SEAT – Junction table for N:M relationship
CREATE TABLE BOOKING_SEAT (
    Booking_ID    INT NOT NULL,
    Seat_ID       INT NOT NULL,
    PRIMARY KEY (Booking_ID, Seat_ID),
    FOREIGN KEY (Booking_ID) REFERENCES BOOKING(Booking_ID)
        ON DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN KEY (Seat_ID)    REFERENCES COACH_SEAT(Seat_ID)
        ON DELETE CASCADE ON UPDATE CASCADE
);

-- 13. PAYMENT Table
CREATE TABLE PAYMENT (
    Payment_ID     INT            PRIMARY KEY AUTO_INCREMENT,
    Booking_ID     INT            NOT NULL UNIQUE,
    Amount         DECIMAL(10,2)  NOT NULL CHECK (Amount > 0),
    Payment_Date   DATE           NOT NULL,
```

```sql
    Payment_Date   DATE           NOT NULL,
    Payment_Status VARCHAR(20)    NOT NULL DEFAULT 'Pending',  -- Pending, Completed, Refunded
    FOREIGN KEY (Booking_ID) REFERENCES BOOKING(Booking_ID)
        ON DELETE CASCADE ON UPDATE CASCADE
);
```

8. Table Creation and Relationships

1. Description of Tables

- **TRAIN:**
  Represents the railway fleet master data. It stores fundamental train details including train number, name, operational category (such as Rajdhani or Express), and total seating capacity.
- **STATION:**
  Maintains the geographic infrastructure network. It stores station identification, station name, city location, and the number of operational platforms.
- **SCHEDULE:**
  Manages timetable instances for each train run, recording arrival times, departure times, and running dates. It depends directly on the Train entity for its existence.
- **TRAIN_STOPS_AT:**
  Defines the intermediate and terminal route halts for each train. It stores the stop sequence order along with station-specific arrival and departure timings.
- **TIMETABLE_EVENT:**
  Serves as an operational aggregation combining a train, a station, and a schedule. It provides a single reference point for ticket booking and crew assignments.
- **EMPLOYEE:**
  The central personnel register storing shared employee attributes, including identification, full name, salary, government identity (Aadhar), contact number, and assigned timetable event.
- **DRIVER:**
  A specialized role under employees, storing locomotive piloting certifications and commercial railway driving license numbers.
- **TC (Ticket Collector):**
  A specialized employee role managing onboard and station ticket inspections, recording official badge numbers and assigned operating zones.
- **MAINTENANCE_STAFF:**
  A specialized employee role managing mechanical and electrical yard maintenance, recording shift timings and technical trade skills.
- **COACH_SEAT:**
  Maintains physical seat inventory inside each train carriage, specifying coach numbers, individual seat numbers, and travel classes.
- **PASSENGER:**
  Stores customer master profiles independently of trips, recording passenger identification, demographic details, phone numbers, and Aadhar numbers.

- **BOOKING:**
  Acts as the primary ticket transaction record, linking a passenger, an operational timetable event, origin and destination stations, travel dates, travel class, and reservation status.
- **BOOKING_SEAT:**
  A junction entity mapping reserved coach seats to specific ticket bookings, allowing multiple seats to be reserved under a single booking.
- **PAYMENT:**
  Records financial transaction details for each booking, including invoice amount, payment date, and transaction status (such as Completed or Refunded).

2. Explanation of Relationships

A. One-to-Many (1:N1:N) Relationships

- **Train to Schedule:** One train operates across multiple scheduled dates, while each schedule belongs to one train.
- **Train to Coach Seat:** One train contains multiple coach seats, while each physical seat belongs to only one train.
- **Passenger to Booking:** A passenger can make multiple bookings over time, but each booking belongs to one primary passenger.
- **Station to Booking:** A station can serve as the starting point or destination for many bookings. Each booking specifies exactly one origin and one destination station.
- **Timetable Event to Booking:** Multiple passengers can book tickets on the same scheduled timetable event.
- **Timetable Event to Employee:** Multiple crew members (drivers, conductors) are assigned to manage a single scheduled timetable event.

B. Many-to-Many (M:NM:N) Relationships

- **Train and Station (via TRAIN_STOPS_AT):** A train stops at multiple stations along its route, and a station hosts halts for many different trains. The junction table stores stop order and halt timings.
- **Booking and Coach Seat (via BOOKING_SEAT):** A single booking can reserve multiple seats for a group, and a physical seat is reserved across multiple bookings over different dates.

C. One-to-One (1:11:1) Relationship

- **Booking and Payment:** Each ticket booking generates exactly one payment transaction, and each payment receipt traces back to one specific booking.

D. Generalization and Specialization (ISA Hierarchy)

- **Employee (Supertype) to Driver, TC, and Maintenance Staff (Subtypes):** All railway personnel inherit shared attributes (Name, Salary, Aadhar, Phone) from the Employee entity, while role-specific attributes (License Number, Badge Number, Shift/Trade Skill) are stored in dedicated subtype tables.

E. Weak Entity Dependencies

- **Schedule on Train:** A schedule cannot exist without an operational train; removing a train removes its schedules.
- **Coach Seat on Train:** A seat is physically located within a train carriage and cannot exist without its parent train.

F. Aggregation

- **Timetable Event:** Unifies the interdependent relationship among Train, Station, and Schedule into a composite entity. Both Bookings and Employees link directly to this aggregated event rather than maintaining multiple disconnected links.

9. Sample Data Insertion:

```sql
USE TrainManagementSystem;

-- TRAIN
INSERT INTO TRAIN (Train_No, Train_Name, Train_Type, Capacity) VALUES
(12951, 'Mumbai Rajdhani Express',   'Rajdhani',   1200),
(12301, 'Howrah Rajdhani Express',   'Rajdhani',   1100),
(12002, 'Bhopal Shatabdi',          'Shatabdi',    800),
(12627, 'Karnataka Express',        'Superfast',   1500),
(12839, 'Chennai Mail',             'Mail',        1800),
(22691, 'Bangalore Rajdhani',       'Rajdhani',    1000),
(12259, 'Sealdah Duronto',          'Duronto',      900),
(12431, 'Trivandrum Rajdhani',      'Rajdhani',    1050),
(11301, 'Udyan Express',            'Express',     1400),
(12049, 'Gatimaan Express',         'Semi-Highspeed', 700);
```

Train_No	Train_Name	Train_Type	Capacity
11301	Udyan Express	Express	1400
12002	Bhopal Shatabdi	Shatabdi	800
12049	Gatimaan Express	Semi-Highspeed	700
12259	Sealdah Duronto	Duronto	900
12301	Howrah Rajdhani Express	Rajdhani	1100
12431	Trivandrum Rajdhani	Rajdhani	1050
12627	Karnataka Express	Superfast	1500
12839	Chennai Mail	Mail	1800
12951	Mumbai Rajdhani Express	Rajdhani	1200
22691	Bangalore Rajdhani	Rajdhani	1000
NULL	NULL	NULL	NULL

```sql
-- STATION
INSERT INTO STATION (Station_ID, Station_Name, City, Platform_No) VALUES
(1,  'New Delhi',               'Delhi',       16),
(2,  'Mumbai Central',          'Mumbai',      12),
(3,  'Howrah Junction',         'Kolkata',     23),
(4,  'Chennai Central',         'Chennai',     17),
(5,  'Bangalore City Junction', 'Bangalore',   10),
(6,  'Bhopal Junction',         'Bhopal',       6),
(7,  'Jaipur Junction',         'Jaipur',       5),
(8,  'Agra Cantt',              'Agra',         5),
(9,  'Trivandrum Central',      'Thiruvananthapuram', 5),
(10, 'Pune Junction',           'Pune',         6),
(11, 'Nagpur Junction',         'Nagpur',       8),
(12, 'Sealdah',                 'Kolkata',      13);
```

Station_ID	Station_Name	City	Platform_No
1	New Delhi	Delhi	16
2	Mumbai Central	Mumbai	12
3	Howrah Junction	Kolkata	23
4	Chennai Central	Chennai	17
5	Bangalore City Junction	Bangalore	10
6	Bhopal Junction	Bhopal	6
7	Jaipur Junction	Jaipur	5
8	Agra Cantt	Agra	5
9	Trivandrum Central	Thiruvananthapuram	5
10	Pune Junction	Pune	6
11	Nagpur Junction	Nagpur	8
12	Sealdah	Kolkata	13
NULL	NULL	NULL	NULL

```sql
-- SCHEDULE
INSERT INTO SCHEDULE (Train_No, Arrival_Time, Departure_Time, Running_Date) VALUES
(12951, '06:00:00', '06:15:00', '2026-10-12'),   -- Schedule_ID = 1
(12951, '16:35:00', '16:35:00', '2026-10-12'),   -- Schedule_ID = 2
(12301, '17:00:00', '17:10:00', '2026-10-12'),   -- Schedule_ID = 3
(12002, '06:15:00', '06:20:00', '2026-10-12'),   -- Schedule_ID = 4
(12627, '21:00:00', '21:15:00', '2026-10-13'),   -- Schedule_ID = 5
(12839, '22:30:00', '22:45:00', '2026-10-13'),   -- Schedule_ID = 6
(22691, '20:50:00', '21:05:00', '2026-10-13'),   -- Schedule_ID = 7
(12259, '12:30:00', '12:45:00', '2026-10-14'),   -- Schedule_ID = 8
(12431, '11:00:00', '11:15:00', '2026-10-14'),   -- Schedule_ID = 9
(11301, '08:10:00', '08:25:00', '2026-10-14'),   -- Schedule_ID = 10
(12049, '08:10:00', '08:10:00', '2026-10-15'),   -- Schedule_ID = 11
(12049, '09:50:00', '09:50:00', '2026-10-15');   -- Schedule_ID = 12
```

Schedule_ID	Train_No	Arrival_Time	Departure_Time	Running_Date
1	12951	06:00:00	06:15:00	2026-10-12
2	12951	16:35:00	16:35:00	2026-10-12
3	12301	17:00:00	17:10:00	2026-10-12
4	12002	06:15:00	06:20:00	2026-10-12
5	12627	21:00:00	21:15:00	2026-10-13
6	12839	22:30:00	22:45:00	2026-10-13
7	22691	20:50:00	21:05:00	2026-10-13
8	12259	12:30:00	12:45:00	2026-10-14
9	12431	11:00:00	11:15:00	2026-10-14
10	11301	08:10:00	08:25:00	2026-10-14
11	12049	08:10:00	08:10:00	2026-10-15
12	12049	09:50:00	09:50:00	2026-10-15

```sql
-- TRAIN_STOPS_AT
INSERT INTO TRAIN_STOPS_AT (Train_No, Station_ID, Stop_Order, Arrival_Time, Departure_Time) VALUES
-- Mumbai Rajdhani: Delhi → Jaipur → Mumbai
(12951, 1,  1, NULL,        '16:35:00'),
(12951, 7,  2, '22:10:00', '22:15:00'),
(12951, 2,  3, '06:00:00',  NULL),
-- Howrah Rajdhani: Delhi → Howrah
(12301, 1,  1, NULL,        '17:10:00'),
(12301, 3,  2, '09:55:00',  NULL),
-- Karnataka Express: Delhi → Agra → Bangalore
(12627, 1,  1, NULL,        '21:15:00'),
(12627, 8,  2, '00:05:00', '00:10:00'),
(12627, 5,  3, '06:40:00',  NULL),
-- Gatimaan Express: Delhi → Agra
(12049, 1,  1, NULL,        '08:10:00'),
(12049, 8,  2, '09:50:00',  NULL);
```

Train_No	Station_ID	Stop_Order	Arrival_Time	Departure_Time
12049	1	1	NULL	08:10:00
12049	8	2	09:50:00	NULL
12301	1	1	NULL	17:10:00
12301	3	2	09:55:00	NULL
12627	1	1	NULL	21:15:00
12627	5	3	06:40:00	NULL
12627	8	2	00:05:00	00:10:00
12951	1	1	NULL	16:35:00
12951	2	3	06:00:00	NULL
12951	7	2	22:10:00	22:15:00
NULL	NULL	NULL	NULL	NULL

```sql
-- TIMETABLE_EVENT — Aggregation of TRAIN + STATION + SCHEDULE
-- ──────────────────────────────────────
INSERT INTO TIMETABLE_EVENT (Event_ID, Train_No, Station_ID, Schedule_ID) VALUES
('EVT-12951-DEL',  12951, 1,  2),   -- Mumbai Rajdhani at Delhi (departure schedule)
('EVT-12951-JPR',  12951, 7,  1),   -- Mumbai Rajdhani at Jaipur
('EVT-12951-MUM',  12951, 2,  1),   -- Mumbai Rajdhani at Mumbai (arrival schedule)
('EVT-12301-DEL',  12301, 1,  3),
('EVT-12301-HWH',  12301, 3,  3),
('EVT-12002-DEL',  12002, 1,  4),
('EVT-12627-DEL',  12627, 1,  5),
('EVT-12627-AGR',  12627, 8,  5),
('EVT-12627-BNG',  12627, 5,  5),
('EVT-12839-DEL',  12839, 1,  6),
('EVT-12839-CHN',  12839, 4,  6),   -- Chennai Mail at Chennai
('EVT-12049-DEL',  12049, 1,  11),  -- Gatimaan Express at Delhi
('EVT-12049-AGR',  12049, 8,  12);  -- Gatimaan Express at Agra
```

Event_ID	Train_No	Station_ID	Schedule_ID
EVT-12002-DEL	12002	1	4
EVT-12049-AGR	12049	8	12
EVT-12049-DEL	12049	1	11
EVT-12301-DEL	12301	1	3
EVT-12301-HWH	12301	3	3
EVT-12627-AGR	12627	8	5
EVT-12627-BNG	12627	5	5
EVT-12627-DEL	12627	1	5
EVT-12839-CHN	12839	4	6
EVT-12839-DEL	12839	1	6
EVT-12951-DEL	12951	1	2
EVT-12951-JPR	12951	7	1
EVT-12951-MUM	12951	2	1

```sql
-- EMPLOYEE (12 employees — 5 Drivers, 5 TCs, 2 Maintenance Staff)
INSERT INTO EMPLOYEE (Employee_ID, Name, Salary, Aadhar_no, Phone_no, Event_ID) VALUES
(1,  'Rajesh Kumar',     65000.00, '234567890123', '9876543210', 'EVT-12951-DEL'),
(2,  'Sunil Sharma',     60000.00, '345678901234', '9876543211', 'EVT-12301-DEL'),
(3,  'Amit Patel',       55000.00, '456789012345', '9876543212', 'EVT-12002-DEL'),
(4,  'Vikram Singh',     62000.00, '567890123456', '9876543213', 'EVT-12627-DEL'),
(5,  'Deepak Verma',     40000.00, '678901234567', '9876543214', 'EVT-12951-DEL'),
(6,  'Manoj Tiwari',     38000.00, '789012345678', '9876543215', 'EVT-12301-DEL'),
(7,  'Ravi Mishra',      42000.00, '890123456789', '9876543216', 'EVT-12002-DEL'),
(8,  'Sanjay Gupta',     39000.00, '901234567890', '9876543217', 'EVT-12627-DEL'),
(9,  'Pradeep Joshi',    58000.00, '123456789012', '9876543218', 'EVT-12839-DEL'),
(10, 'Ankit Chauhan',    37000.00, '112233445566', '9876543219', 'EVT-12839-DEL'),
(11, 'Ramesh Yadav',     35000.00, '223344556677', '9876543220', 'EVT-12951-MUM'),
(12, 'Suresh Pandey',    36000.00, '334455667788', '9876543221', 'EVT-12301-HWH');
```

Employee_ID	Name	Salary	Aadhar_no	Phone_no	Event_ID
1	Rajesh Kumar	65000.00	234567890123	9876543210	EVT-12951-DEL
2	Sunil Sharma	60000.00	345678901234	9876543211	EVT-12301-DEL
3	Amit Patel	55000.00	456789012345	9876543212	EVT-12002-DEL
4	Vikram Singh	62000.00	567890123456	9876543213	EVT-12627-DEL
5	Deepak Verma	40000.00	678901234567	9876543214	EVT-12951-DEL
6	Manoj Tiwari	38000.00	789012345678	9876543215	EVT-12301-DEL
7	Ravi Mishra	42000.00	890123456789	9876543216	EVT-12002-DEL
8	Sanjay Gupta	39000.00	901234567890	9876543217	EVT-12627-DEL
9	Pradeep Joshi	58000.00	123456789012	9876543218	EVT-12839-DEL
10	Ankit Chauhan	37000.00	112233445566	9876543219	EVT-12839-DEL
11	Ramesh Yadav	35000.00	223344556677	9876543220	EVT-12951-MUM
12	Suresh Pandey	36000.00	334455667788	9876543221	EVT-12301-HWH

```sql
-- DRIVER
INSERT INTO DRIVER (Employee_ID, License_No) VALUES
(1, 'DL-2020-RLY-001'),
(2, 'DL-2019-RLY-045'),
(3, 'DL-2021-RLY-112'),
(4, 'DL-2018-RLY-078'),
(9, 'DL-2022-RLY-203');
```

Employee_ID	License_No
4	DL-2018-RLY-078
2	DL-2019-RLY-045
1	DL-2020-RLY-001
3	DL-2021-RLY-112
9	DL-2022-RLY-203

```
112     -- TC
113 •   INSERT INTO TC (Employee_ID, Badge_No, Assigned_Zone) VALUES
114     (5,  'TC-MUM-0501', 'Western Zone'),
115     (6,  'TC-HWH-0602', 'Eastern Zone'),
116     (7,  'TC-BPL-0703', 'West Central Zone'),
117     (8,  'TC-BNG-0804', 'South Western Zone'),
118     (10, 'TC-CHN-1005', 'Southern Zone');
```

Employee_ID	Badge_No	Assigned_Zone
5	TC-MUM-0501	Western Zone
6	TC-HWH-0602	Eastern Zone
7	TC-BPL-0703	West Central Zone
8	TC-BNG-0804	South Western Zone
10	TC-CHN-1005	Southern Zone
NULL	NULL	NULL

```
121     -- MAINTENANCE_STAFF
122 •   INSERT INTO MAINTENANCE_STAFF (Employee_ID, Shift_Type, Trade_Skill) VALUES
123     (11, 'Night',       'Electrical'),
124     (12, 'Day',  'Mechanical');
```

Employee_ID	Shift_Type	Trade_Skill
11	Night	Electrical
12	Day	Mechanical
NULL	NULL	NULL

```
    INSERT INTO COACH_SEAT (Train_No, Coach_No, Seat_No, Class) VALUES
130 -- Mumbai Rajdhani (Seat_ID 1–10)
131 (12951, 'A1', 1,  '1AC'),
132 (12951, 'A1', 2,  '1AC'),
133 (12951, 'A1', 3,  '1AC'),
134 (12951, 'B1', 1,  '2AC'),
135 (12951, 'B1', 2,  '2AC'),
136 (12951, 'B1', 3,  '2AC'),
137 (12951, 'B2', 1,  '2AC'),
138 (12951, 'B2', 2,  '2AC'),
139 (12951, 'S1', 1,  'SL'),
140 (12951, 'S1', 2,  'SL'),
141 -- Howrah Rajdhani (Seat_ID 11–18)
142 (12301, 'A1', 1,  '1AC'),
143 (12301, 'A1', 2,  '1AC'),
144 (12301, 'B1', 1,  '2AC'),
145 (12301, 'B1', 2,  '2AC'),
146 (12301, 'B1', 3,  '2AC'),
147 (12301, 'S1', 1,  'SL'),
148 (12301, 'S1', 2,  'SL'),
149 (12301, 'S1', 3,  'SL'),
150 -- Karnataka Express (Seat_ID 19–25)
151 (12627, 'B1', 1,  '2AC'),
152 (12627, 'B1', 2,  '2AC'),
153 (12627, 'S1', 1,  'SL'),
154 (12627, 'S1', 2,  'SL'),
155 (12627, 'S1', 3,  'SL'),
156 (12627, 'G1', 1,  'GEN'),
157 (12627, 'G1', 2,  'GEN');
```

Seat_ID	Train_No	Coach_No	Seat_No	Class
1	12951	A1	1	1AC
2	12951	A1	2	1AC
3	12951	A1	3	1AC
4	12951	B1	1	2AC
5	12951	B1	2	2AC
6	12951	B1	3	2AC
7	12951	B2	1	2AC
8	12951	B2	2	2AC
9	12951	S1	1	SL
10	12951	S1	2	SL
11	12301	A1	1	1AC
12	12301	A1	2	1AC
13	12301	B1	1	2AC
14	12301	B1	2	2AC
15	12301	B1	3	2AC
16	12301	S1	1	SL
17	12301	S1	2	SL
18	12301	S1	3	SL
19	12627	B1	1	2AC
20	12627	B1	2	2AC
21	12627	S1	1	SL
22	12627	S1	2	SL
23	12627	S1	3	SL
24	12627	G1	1	GEN
25	12627	G1	2	GEN
NULL	NULL	NULL	NULL	NULL

```
161    -- PASSENGER
162 •  INSERT INTO PASSENGER (Passenger_ID, Name, Age, Gender, Phone_no, Aadhar_no) VALUES
163    (1,  'Aarav Mehta',      28, 'Male',   '8001234501', '100020003001'),
164    (2,  'Priya Sharma',     35, 'Female', '8001234502', '100020003002'),
165    (3,  'Rohan Desai',      42, 'Male',   '8001234503', '100020003003'),
166    (4,  'Sneha Iyer',       25, 'Female', '8001234504', '100020003004'),
167    (5,  'Karthik Nair',     55, 'Male',   '8001234505', '100020003005'),
168    (6,  'Ananya Reddy',     30, 'Female', '8001234506', '100020003006'),
169    (7,  'Vikash Yadav',     19, 'Male',   '8001234507', '100020003007'),
170    (8,  'Meera Joshi',      60, 'Female', '8001234508', '100020003008'),
171    (9,  'Arjun Kapoor',     33, 'Male',   '8001234509', '100020003009'),
172    (10, 'Divya Pillai',     22, 'Female', '8001234510', '100020003010');
```

Passenger_ID	Name	Age	Gender	Phone_no	Aadhar_no
1	Aarav Mehta	28	Male	8001234501	100020003001
2	Priya Sharma	35	Female	8001234502	100020003002
3	Rohan Desai	42	Male	8001234503	100020003003
4	Sneha Iyer	25	Female	8001234504	100020003004
5	Karthik Nair	55	Male	8001234505	100020003005
6	Ananya Reddy	30	Female	8001234506	100020003006
7	Vikash Yadav	19	Male	8001234507	100020003007
8	Meera Joshi	60	Female	8001234508	100020003008
9	Arjun Kapoor	33	Male	8001234509	100020003009
10	Divya Pillai	22	Female	8001234510	100020003010
NULL	NULL	NULL	NULL	NULL	NULL

```sql
-- BOOKING
INSERT INTO BOOKING (Booking_ID, Passenger_ID, Event_ID, Start_Station,
End_Station, Booking_Date, Journey_Date, Status, Class) VALUES
(1,  1,  'EVT-12951-DEL', 1, 2,  '2026-10-01', '2026-10-12', 'Confirmed', '1AC'),
(2,  2,  'EVT-12951-DEL', 1, 2,  '2026-10-02', '2026-10-12', 'Confirmed', '2AC'),
(3,  3,  'EVT-12301-DEL', 1, 3,  '2026-10-03', '2026-10-12', 'Confirmed', '1AC'),
(4,  4,  'EVT-12627-DEL', 1, 5,  '2026-10-04', '2026-10-13', 'Waiting',   '2AC'),
(5,  5,  'EVT-12839-DEL', 1, 4,  '2026-10-05', '2026-10-13', 'Confirmed', 'SL'),
(6,  6,  'EVT-12002-DEL', 1, 6,  '2026-10-06', '2026-10-12', 'Cancelled', '2AC'),
(7,  7,  'EVT-12049-DEL', 1, 8,  '2026-10-07', '2026-10-15', 'Confirmed', 'GEN'),
(8,  8,  'EVT-12951-DEL', 1, 7,  '2026-10-08', '2026-10-12', 'Confirmed', 'SL'),
(9,  9,  'EVT-12301-DEL', 1, 3,  '2026-10-09', '2026-10-12', 'Confirmed', '2AC'),
(10, 10, 'EVT-12627-DEL', 1, 5,  '2026-10-10', '2026-10-13', 'Confirmed', 'SL');

-- BOOKING_SEAT
INSERT INTO BOOKING_SEAT (Booking_ID, Seat_ID) VALUES
(1, 1),
(2, 4),
(2, 5),
(3, 11),
(5, 9),
(7, 24),
(8, 10),
(9, 13),
(10, 21);
```

Passenger_ID	Name	Age	Gender	Phone_no	Aadhar_no
1	Aarav Mehta	28	Male	8001234501	100020003001
2	Priya Sharma	35	Female	8001234502	100020003002
3	Rohan Desai	42	Male	8001234503	100020003003
4	Sneha Iyer	25	Female	8001234504	100020003004
5	Karthik Nair	55	Male	8001234505	100020003005
6	Ananya Reddy	30	Female	8001234506	100020003006
7	Vikash Yadav	19	Male	8001234507	100020003007
8	Meera Joshi	60	Female	8001234508	100020003008
9	Arjun Kapoor	33	Male	8001234509	100020003009
10	Divya Pillai	22	Female	8001234510	100020003010
NULL	NULL	NULL	NULL	NULL	NULL

```
203     -- PAYMENT
204 •   INSERT INTO PAYMENT (Booking_ID, Amount, Payment_Date, Payment_Status) VALUES
205     (1,  4500.00,  '2026-10-01', 'Completed'),
206     (2,  2800.00,  '2026-10-02', 'Completed'),
207     (3,  4200.00,  '2026-10-03', 'Completed'),
208     (4,  2600.00,  '2026-10-04', 'Pending'),
209     (5,   850.00,  '2026-10-05', 'Completed'),
210     (6,  2800.00,  '2026-10-06', 'Refunded'),
211     (7,   250.00,  '2026-10-07', 'Completed'),
212     (8,   650.00,  '2026-10-08', 'Completed'),
213     (9,  3200.00,  '2026-10-09', 'Completed'),
214     (10,  750.00,  '2026-10-10', 'Completed');
```

Passenger_ID	Name	Age	Gender	Phone_no	Aadhar_no
1	Aarav Mehta	28	Male	8001234501	100020003001
2	Priya Sharma	35	Female	8001234502	100020003002
3	Rohan Desai	42	Male	8001234503	100020003003
4	Sneha Iyer	25	Female	8001234504	100020003004
5	Karthik Nair	55	Male	8001234505	100020003005
6	Ananya Reddy	30	Female	8001234506	100020003006
7	Vikash Yadav	19	Male	8001234507	100020003007
8	Meera Joshi	60	Female	8001234508	100020003008
9	Arjun Kapoor	33	Male	8001234509	100020003009
10	Divya Pillai	22	Female	8001234510	100020003010
NULL	NULL	NULL	NULL	NULL	NULL

11. Conclusion

The Train Management System database has been successfully designed and implemented using MySQL. The project demonstrates ER Diagram , logical design (Relational Schema) and physical implementation (SQL). The database is normalized, maintains data integrity, and supports efficient querying for train management operations including booking, scheduling, employee management, and revenue tracking.